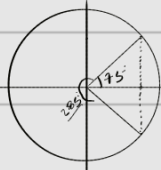


1) a) $\cos 75 \rightarrow \cos(45+30) \rightarrow \cos 45 \cdot \cos 30 + \sin 45 \cdot \sin 30 \rightarrow \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} \left\} \frac{\sqrt{6} + \sqrt{2}}{4}\right.$
 b) $\sin 705 \rightarrow \sin(60+45) \rightarrow \sin 60 \cdot \cos 45 + \sin 45 \cdot \cos 60 \rightarrow \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} \left\} \frac{\sqrt{6} + \sqrt{2}}{4}\right.$
 c) $\tan 75 \rightarrow \tan(30+45) \rightarrow \frac{\tan 30 + \tan 45}{1 - \tan 30 \cdot \tan 45} \rightarrow \frac{\frac{\sqrt{3}}{3} + 1}{1 - \frac{\sqrt{3}}{3}} \left\} \frac{\sqrt{3} + 3}{3 - \sqrt{3}} \left\} \frac{\sqrt{3} + 3}{3 - \sqrt{3}} \cdot \frac{(3 + \sqrt{3})}{(3 + \sqrt{3})} \right\} 2 + \sqrt{3}\right.$

d)  $\sec 285 \rightarrow \frac{1}{\cos 75} \rightarrow \frac{1}{\cos 45 \cdot \cos 30 - \sin 45 \cdot \sin 30} \left\} \frac{1}{\frac{\sqrt{6} + \sqrt{2}}{4}} \right\} \textcircled{I}$
 $\cos(45+30) \quad \textcircled{I} \frac{4}{\sqrt{6} + \sqrt{2}} \cdot \frac{(\sqrt{6} - \sqrt{2})}{(\sqrt{6} - \sqrt{2})} \left\} \sqrt{6} + \sqrt{2} //$

	30°	45°	60°
Senθ	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
Cosθ	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
Tgθ	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$

e) $\cotg 705 \rightarrow \frac{1}{\tan 75} \rightarrow \frac{1}{\tan(45+30)} \rightarrow \frac{1}{\frac{\tan 45 + \tan 30}{1 - \tan 45 \cdot \tan 30}} \left\} \frac{1 - \frac{\sqrt{3}}{3}}{\frac{3 - \sqrt{3}}{3}} \left\} \frac{3 - \sqrt{3}}{3 - \sqrt{3}} \cdot \frac{3}{3 + \sqrt{3}} \left\} \frac{3 - \sqrt{3}}{3 + \sqrt{3}} \left\} \frac{1}{2 - \sqrt{3}} \left\} -2 - \sqrt{3} //$

f) $\sec 255 \rightarrow \frac{1}{\cos 75} \rightarrow \frac{1}{\cos 60 \cdot \cos 15 - \sin 60 \cdot \sin 15} \rightarrow \frac{1}{\frac{\sqrt{2} - \sqrt{6}}{4}} \rightarrow (-\sqrt{2} - \sqrt{6}) //$

g) $\sec 75 \rightarrow \frac{1}{\sin 75} \rightarrow \frac{1}{\sin(45+30)} \rightarrow \frac{1}{\sin 45 \cdot \cos 30 + \cos 45 \cdot \sin 30} \rightarrow \sqrt{6} + \sqrt{2} //$

h) $\sin 705 - \cos 75 \rightarrow \frac{\sqrt{6} + \sqrt{2}}{4} - \frac{\sqrt{6} - \sqrt{2}}{4} \left\} \frac{\sqrt{6} + \sqrt{2} - (\sqrt{6} - \sqrt{2})}{4} \left\} \frac{\sqrt{6} + \sqrt{2} + \sqrt{2} + \sqrt{2}}{4} \left\} \frac{2\sqrt{2}}{4} \left\} \frac{\sqrt{2}}{2} //$

2) $(\sin x + \cos x)^2 - \sin 2x \rightarrow (\sin x)^2 + 2 \sin x \cdot \cos x + (\cos x)^2 - \sin(2x) \rightarrow (\sin x)^2 + 2 \sin x \cdot \cos x + (\cos x)^2 - 2 \sin x \cdot \cos x = \textcircled{I}$
 $\sin(x+x) \quad \textcircled{I} (\sin x)^2 + (\cos x)^2 //$

3) $\cos^2 + \sin^2 = 1 \rightarrow \sin^2 x = 1 - \cos^2 x$

$\cos(x+x) \rightarrow \cos x \cdot \cos x - \sin x \cdot \sin x \rightarrow \cos^2 x - \sin^2 x = \frac{1}{2} \rightarrow \cos^2 x - 1 + \cos^2 x = \frac{1}{2} \rightarrow 2\cos^2 x - 1 = \frac{1}{2} \rightarrow \cos x = \frac{\sqrt{3}}{2}$
 $\tan^2(x) + \sec^2(x) \rightarrow \frac{\sin^2(x)}{\cos^2(x)} + \frac{1}{\cos^2(x)} \rightarrow \frac{\frac{1}{4}}{\frac{3}{4}} + \frac{1}{\frac{3}{4}} \left\} \frac{\frac{1}{4}}{\frac{3}{4}} + \frac{1}{\frac{3}{4}} \left\} \frac{\frac{1}{4} + 1}{\frac{3}{4}} \left\} \frac{\frac{5}{4}}{\frac{3}{4}} \left\} \frac{5}{3} //$ $x = 30^\circ \quad \sin x = \frac{1}{2}$

4) $\cos^2 x + \sin^2 x = 1 \rightarrow \cos^2 x = 1 - \sin^2 x \quad \cos 2x - \sin 2x = \frac{1}{2} \rightarrow (\cos x - \sin x)^2 = \left(\frac{1}{2}\right)^2 \rightarrow \cos^2 x - 2 \sin x \cdot \cos x + \sin^2 x = \frac{1}{4} \rightarrow \textcircled{I}$

$\sin 2x \rightarrow \sin x \cdot \cos x + \sin x \cdot \cos x \rightarrow 2 \sin x \cdot \cos x \quad \textcircled{I} \cos^2 + \sin^2 - \sin(2x) = \frac{1}{4} \rightarrow \sin 2x = 1 - \frac{1}{4} \rightarrow \sin 2x = \frac{3}{4} //$

5) $\sin x = \frac{3}{5} \quad \cos y = \frac{5}{13} \quad \cos(x+y) \rightarrow \cos x \cdot \cos y - \sin x \cdot \sin y \rightarrow \frac{4}{5} \cdot \frac{5}{13} - \frac{3}{5} \cdot \frac{12}{13} \rightarrow \frac{20 + 36}{5 \cdot 13} \quad \frac{56}{65} //$
 $\cos x = \frac{4}{5} \quad \sin y = \frac{12}{13}$

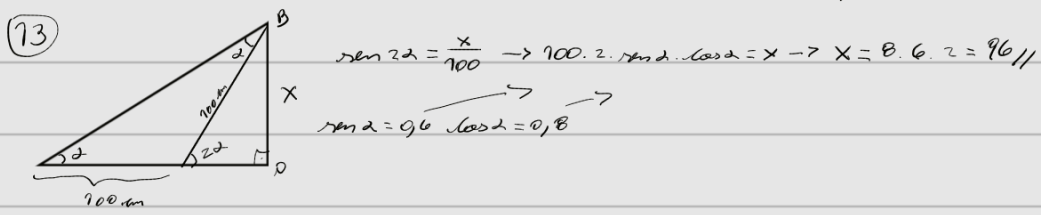
7) $\tan A = \frac{3}{4} \quad \tan(A-B) \rightarrow \frac{\tan A - \tan B}{1 + \tan A \cdot \tan B} \rightarrow \frac{-6}{17} //$
 $\tan B = -\frac{9}{3}$

6) $\tan a = 2 \quad \tan b = 1$
 $\tan(a-b) \rightarrow \frac{2-1}{1+2} \rightarrow \frac{1}{3} //$

9) $\tan 75^\circ = 2 + \sqrt{3} \quad \tan(75-60) \rightarrow \frac{\tan 75 - \tan 60}{1 + \tan 75 \cdot \tan 60} \rightarrow \frac{2 - \sqrt{3} + \sqrt{3}}{1 + (2\sqrt{3} - 3)\sqrt{3}} \rightarrow \frac{2}{4 + 2\sqrt{3}} \rightarrow 2 - \sqrt{3} //$
 $\tan 60^\circ = \sqrt{3}$

10) $\cos 72^\circ - \cos^2 36^\circ \rightarrow \cos(36+36) - \cos^2 36^\circ \rightarrow \cos^2 36^\circ - \sin^2 36^\circ - \cos^2 36^\circ \rightarrow -\sin^2 36^\circ //$

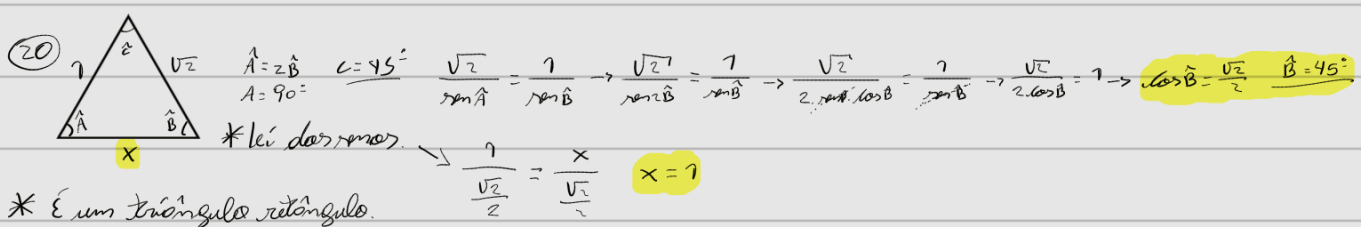
12) $\sin x - \cos x = \sqrt{0}$ $\sin 2x \rightarrow 2 \sin x \cdot \cos x$
 $\sin^2 x + \cos^2 x = 1$ $\left(\sin x - \cos x = \sqrt{0} \rightarrow (\sin x - \cos x)^2 = (\sqrt{0})^2 \rightarrow \sin^2 x - 2 \sin x \cdot \cos x + \cos^2 x = 0 \right)$ (I)
 $\sin^2 x = 1 - \cos^2 x$ (II) $1 - \sin 2x = 0 \rightarrow \sin 2x = -0 + 1 //$



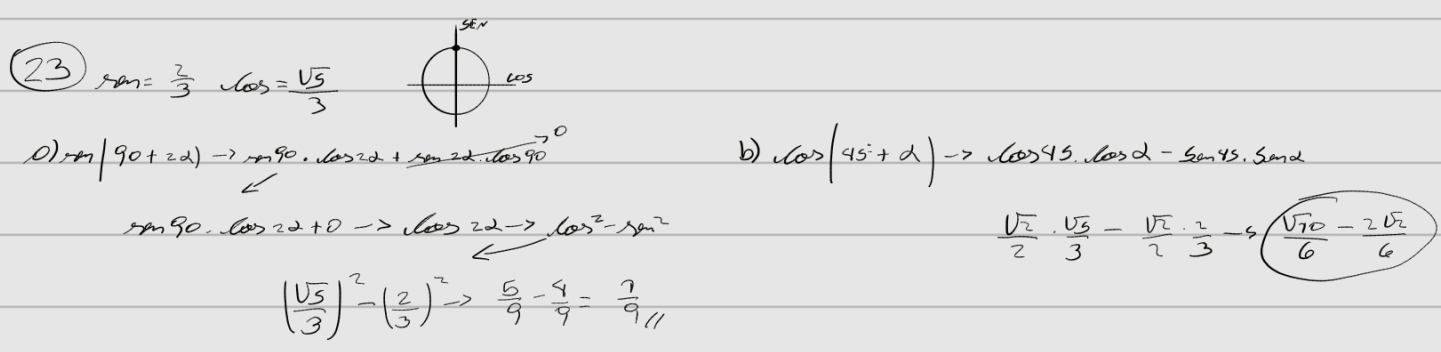
15) $\cos = \frac{3}{5}$ $\sin = -\frac{4}{5}$ $\sin 3x \rightarrow \sin(2x+x) \rightarrow \sin 2x \cdot \cos x + \cos 2x \cdot \sin x \rightarrow 2 \sin x \cdot \cos x \cdot \cos x + (\cos^2 x - \sin^2 x) \cdot \sin x \rightarrow$ (I)
 $\sin 2x = 2 \sin x \cdot \cos x$ (II) $3 \sin x \cdot \cos^2 x - \sin^3 x \rightarrow 3 \sin x \cdot (1 - \sin^2 x) - \sin^3 x \rightarrow 3 \sin x - 4 \sin^3 x$
 $\cos^2 x = 1 - \sin^2 x$ Então $\rightarrow \sin 3x = 3 \sin x - 4 \sin^3 x \therefore \sin 3x = \frac{-44}{125} //$

16) $\sin x = \frac{12}{13}$ $\cos x = -\frac{5}{13}$ $\cos 3x \rightarrow \cos(2x+x) \rightarrow \cos 2x \cdot \cos x - \sin 2x \cdot \sin x \rightarrow (2 \cos^2 x - 1) \cdot \cos x - (2 \sin x \cdot \cos x) \cdot \sin x \rightarrow$ (I)
 $\sin^2 x = 1 - \cos^2 x$ (II) $2 \cos^3 x - \cos x - 2 \cos x \sin^2 x \rightarrow 2 \cos^3 x - \cos x - 2 \cos x (1 - \cos^2 x) \rightarrow$ (II)
 $\cos 3x = 4 \cos^3 x - 3 \cos x$
 $\cos 3x = \frac{2036}{2797}$

17) $\sec = \frac{4}{3} \rightarrow \frac{1}{\cos} = \frac{4}{3} \rightarrow \cos x = \frac{3}{4}$ e $\sin x = \frac{\sqrt{7}}{4} \rightarrow \text{tg} = \frac{\sqrt{7}}{3}$
 $\text{tg} 3x \rightarrow \text{tg}(2x+x) \rightarrow \frac{\text{tg} 2x + \text{tg} x}{1 - \text{tg} 2x \cdot \text{tg} x} \rightarrow \frac{\frac{2 \text{tg} x + \text{tg} x}{1 - \text{tg}^2 x}}{1 - \frac{2 \text{tg}^2 x}{1 - \text{tg}^2 x}} \rightarrow \frac{\frac{3 \text{tg} x - \text{tg}^3 x}{1 - \text{tg}^2 x}}{\frac{1 - \text{tg}^2 x - 2 \text{tg}^2 x}{1 - \text{tg}^2 x}} \rightarrow \frac{3 \text{tg} x - \text{tg}^3 x}{1 - 3 \text{tg}^2 x} \rightarrow$ (I)
 $\text{tg} = \frac{\sqrt{7}}{3}$ (II) * Fazendo o substituição: $\text{tg} 3x = \frac{3 \text{tg} x - \text{tg}^3 x}{1 - 3 \text{tg}^2 x} \rightarrow \frac{3 \cdot \frac{\sqrt{7}}{3} - \left(\frac{\sqrt{7}}{3}\right)^3}{1 - 3 \cdot \left(\frac{\sqrt{7}}{3}\right)^2} \rightarrow \frac{\frac{20\sqrt{7}}{27}}{-\frac{4}{3}} \rightarrow \text{tg} 3x = -\frac{5\sqrt{7}}{9}$

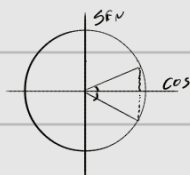


21) $\text{tg} x + \cotg x = 3 \rightarrow \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = 3 \rightarrow \frac{\sin^2 x + \cos^2 x}{\sin x \cdot \cos x} = 3 \rightarrow \frac{1}{\sin x \cdot \cos x} = 3 \rightarrow \sin x \cdot \cos x = \frac{1}{3}$
 $\sin 2x = 2 \sin x \cdot \cos x \rightarrow \sin 2x = \frac{2}{3}$



25) $\sin^2 x + \cos^2 x = 1 \rightarrow \sin x = \sqrt{1 - \cos^2 x}$

a) $\sqrt{3} \cos x + 1 \sin x = 1 \rightarrow \frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x = \frac{1}{2} \rightarrow \cos \left| \frac{\pi}{6} - x \right| = \frac{1}{2}$

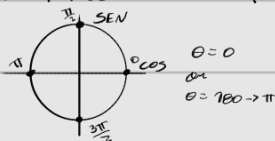


$\frac{\pi}{6} - x = \theta$ e $\theta = \pm \frac{\pi}{3}$ ou seja $\rightarrow \frac{\pi}{3} + 2k\pi = \frac{\pi}{6} - x \rightarrow x = -\frac{\pi}{6} - 2k\pi$

$\left(-\frac{\pi}{3} + 2k\pi = \frac{\pi}{6} - x \right) \rightarrow x = \frac{\pi}{2} - 2k\pi$

* Soluções

b) $\sin x + \cos x = 1 \rightarrow (\sin x + \cos x)^2 = 1^2 \rightarrow \sin^2 x + 2 \sin x \cos x + \cos^2 x = 1 \rightarrow \sin x \cos x = 0$



$\sin x = 0$ ou $\cos x = 0$

$\hookrightarrow x = k\pi$

$\hookrightarrow x = \frac{\pi}{2} + k\pi$

* Vamos validar na expressão original. $\oplus$

$\sin x + \cos x = 1$

* Portanto, note-se que a redução para a equação é:

Ⓘ) para $\pi \rightarrow 0 - 1 = -1 \rightarrow$ Não Satisfaz

para $2\pi \rightarrow 0 + 1 = 1 \rightarrow$ Satisfaz

para $\frac{3\pi}{2} \rightarrow -1 + 0 = -1 \rightarrow$ Não Satisfaz

para $\frac{\pi}{2} \rightarrow 1 + 0 = 1 \rightarrow$ Satisfaz

$x = 2k\pi$

$x = \frac{\pi}{2} + 2k\pi$

26) b) $\frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x} \rightarrow \frac{3 \sin x - 4 \sin^3 x}{\sin x} - \frac{-3 \cos x + 4 \cos^3 x}{\cos x} \rightarrow \frac{\sin(3 - 4 \sin^2 x)}{\sin} - \frac{\cos(-3 + 4 \cos^2 x)}{\cos}$

$\rightarrow 3 - 4 \sin^2 x + 3 - 4 \cos^2 x \rightarrow -4(\cos^2 x + \sin^2 x) + 6 \rightarrow 2$

c) $\frac{(1 - \sqrt{2} \sin x)(1 + \sqrt{2} \sin x) + (1 + \sqrt{2} \cos x)(1 - \sqrt{2} \cos x)}{(1 + \sqrt{2} \cos x)(1 + \sqrt{2} \sin x)} \rightarrow \frac{2 - 2 \sin^2 x - 2 \cos^2 x}{(1 + \sqrt{2} \cos x)(1 + \sqrt{2} \sin x)} \rightarrow \frac{-2(-1 + \sin^2 x + \cos^2 x)}{(1 + \sqrt{2} \cos x)(1 + \sqrt{2} \sin x)}$

* Nesta parte vai sempre dar 0, pois $\sin^2 + \cos^2 = 1$ e $(-1 + 1) \cdot 2 = 0$

* O resultado é 0